Problem A
Shout!
Time limit: 3 seconds

The ship has sunk, and your only hope is to get the attention of those back on land! To do this, your plan is to shout a message at them, possibly many times. No more than 5 times though: any more than that, and you'd get too tired. Can you shout loud and clear, but not too much?

Input

Input consists of 2 lines.

The first line of input contains a string of alphanumeric characters, consisting of the message to shout. You are guaranteed that this string is nonempty and at most 100 characters long.

The second line of input contains a single integer n $(1 \leq n \leq 100)$, the number of times you're instructed to shout.

Note: Constraints such as $(1 \leq n \leq 100)$ *given in the input are* ***guarantees****. That is, all inputs you are tested on will satisfy these constraints. This applies for all problems on this contest.*

Output

If instructed to shout 5 times or fewer, print the message to be shouted n times, each on a separate line. Otherwise, print it 5 times, each on a separate line, followed by the string `I'm tired` also on a separate line.

Sample Input 1	Sample Output 1
`help` `3`	`help` `help` `help`

Sample Input 2	Sample Output 2

`sh0ut`
`7`
	`sh0ut`
	`sh0ut`
	`sh0ut`
	`sh0ut`
	`sh0ut`
	`I'm tired`

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Problem B
Battleship
Time limit: 3 seconds

Admiral Clement Maximilian of the High Seas Patrol Command has found himself in a dire situation: His fleet is rapidly approaching a vast armada of enemy ships, spread out over a large section of ocean. Luckily, the Admiral has practiced for this very scenario, and he knows that if he strikes the ships in the right places, he can sink them all before they have a chance to retaliate! He has already successfully landed one strike on each ship, but he needs your help to finish the job. He has limited ammunition and can't afford to make many mistakes. Can you help the Admiral sink all the enemy ships without making too many incorrect guesses?

The Admiral uses an $n \times n$ grid to represent the ocean. Each cell of the grid is either empty or contains a part of a ship. All the enemy ships are the same length, l. The ships are located on the grid according to the following rules:

- Each ship occupies a contiguous sequence of l cells in either a horizontal or vertical direction.
- No two ships are adjacent to each other. This means that any two cells that share an edge cannot be occupied by different ships.

At each step, the Admiral will communicate to you the current state of the grid. The grid is represented by n lines of text, each containing a string of length n, composed of the characters `# . ?`. The c-th character of the r-th line of text represents the state of the cell in row r and column c:

- A `#` represents a cell that has been attacked (either initially by the Admiral, or later by you) and contains a ship.
- A `.` represents a cell that you have attacked that does not contain a ship. Whenever you attack a cell that does not contain a ship, this counts as a *miss*.
- A `?` represents a cell that has not been attacked yet, either by you or the Admiral.

Given one cell on each ship that the Admiral has already struck, make attacks to sink all the ships while making no more than $3k$ misses.

Interaction

This is an interactive problem.

The first line of input contains a single integer n ($1 \leq n \leq 40$), the size of the grid.

The second line of input contains a single integer k ($1 \leq k \leq n^2$), the number of ships.

The third line of input contains a single integer l ($2 \leq l \leq n$), the length of each ship.

The next n lines of input describe the initial state of the grid, as described above. All cells of this initial grid are either a `#` (attacked by the Admiral) or a `?` (not yet attacked). You are guaranteed that exactly one cell of each ship is a `#`.

To make an attack, output a single line of the format `? r c`, where r and c denote the row and column of the cell to attack. Since this is an interactive problem, you must flush the output stream after each line of output. This can be done with `fflush(stdout)` in C++, `System.out.flush()` in Java, or `sys.stdout.flush()` in Python.

After each attack, you will receive n lines of input, denoting the new state of the grid. That is, the character in row r and column c will be updated with the new status of that cell: either `#` if the attack hit a ship, or `.` if it missed.

You may make a maximum of $3k$ misses. If your program receives its $3k + 1$-st miss, it will be terminated and you will receive a verdict of `WRONG ANSWER`. You will also receive a `WRONG ANSWER` verdict if you make a guess out-of-bounds or in an incorrect format.

Once you have sunk all the ships, output a single line containing the character `!`. A ship is considered sunk when all the cells it occupies have been attacked (either by you, or by the Admiral's initial attacks). You will receive an `ACCEPTED` verdict if you have in fact sunk all the ships, and a `WRONG ANSWER` verdict otherwise.

Sample Interaction 1

Read	Write
5 2 3 ????? ????# ????? ?#??? ?????	
	? 1 5
????# ????# ????? ?#??? ?????	
	? 1 1
.???# ????# ????? ?#??? ?????	
	> 3 5
.???# ????# ????# ?#??? ?????	
	? 4 4
.???# ????# ????# ?#?#? ?????	
	? 4 3
.???# ????# ????# ?###? ?????	
	!

Problem C
Altimeter
Time limit: 5 seconds

You've been exploring the depths of the oceans, from the surface-level Epipelagic zone to the deep-down Abyssal zone. You've mapped out all the zones that are relevant to your work, and now you need a fast way to determine which zone you're in. Each zone represents an interval of depths under the sea, with each depth from 0 to the seafloor corresponding to exactly one zone. Given a list of zones and their boundaries, can you efficiently answer queries about which zone a given depth lies in?

Input

The first line contains an integer n $(1 \le n \le 2 \cdot 10^5)$,.

Following it are n lines of input. The i-th of these n lines contains a string s_i, followed by two integers a_i and b_i $(0 \le a_i, b_i < 10^9)$, space-separated. The string s_i is a string of at most 16 Latin letters of any case, representing the name of zone i, and the integers a_i and b_i represent the minimum and maximum depths of this zone.
You are guaranteed that the first zone extends up to the surface (i.e. $a_1 = 0$), and that zones are consecutive and given in order (i.e. $b_{i-1} = a_i$ for all $2 \le i \le n$).

The next line contains an integer q $(1 \le q \le 2 \cdot 10^5)$, the number of queries.

Each of the next q lines contains an integer q_i, representing the depth that is being queried. You are guaranteed that q_i does not lie on the boundary between two zones (i.e. $q_i \ne a_j$ and $q_i \ne b_j$ for all j), and that q_i lies between the minimum and maximum depth of all the zones (i.e. $0 < q_i < b_n$).

Output

Output q lines, where the i-th line of output is the name of the zone containing depth q_i.

Sample Input 1	Sample Output 1
4 Epipelagic 0 100 Mesopelagic 100 1000 Bathypelagic 1000 3000 Abyssal 3000 10000 6 500 30 9000 3001 2999 1	Mesopelagic Epipelagic Abyssal Abyssal Bathypelagic Epipelagic

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Problem D
Reasonable Offers
Time limit: 5 seconds

Your scuba diving training is finally paying off! Several deep-sea diving teams have reached out to you with job offers, with generous compensation and benefits. In fact, there are so many offers that you're not sure how to even process them all! It looks like you need to narrow down your options, somehow...

After thinking for a bit, you've decided to classify your job offers through two metrics: *salary* and *vacation days*, both of which are integers between 0 and 10^9 inclusive. You consider a job offer to be *reasonable* if every other offer has either a lower salary or fewer vacation days. Given a set of job offers, can you determine how many are reasonable?

Input

The first line of input contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$), the number of job offers you have received.

n lines of input follow. The i-th of these lines of input contains two integers s_i, v_i, space-separated, indicating that the i-th job offer you've received has a salary of s_i, and v_i vacation days.

You are guaranteed that no two offers have both the same salary and the same number of vacation days. That is, if $i \neq j$, then either $s_i \neq s_j$ or $v_i \neq v_j$ (or both).

Output

Output a single integer, the number of offers that are reasonable.

Sample Input 1	Sample Output 1
5 0 1000000000 40000 10 100000 0 30000 25 50000 15	4

Sample Input 2	Sample Output 2
4 60000 0 60000 15 30000 0 30000 15	1

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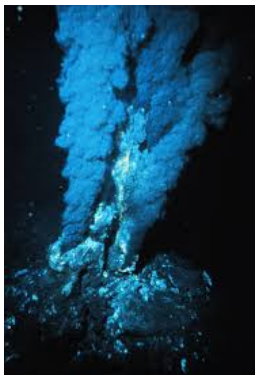
Problem E
Hydrothermal Vents (Div. 2)
Time limit: 2 seconds



You've been exploring the deep sea, and you have one final task before returning to your base: collecting a sample from one of the hydrothermal vents scattered throughout the terrain. Your crew has provided you with a handy map of the terrain, as well as instructions for your final mission.

The seafloor terrain is represented as a rectangular grid, with cells of the following types:

`A`	Start (your submarine)
`B`	Base (your destination)
`^`	Hydrothermal vent
`#`	Impassable rock
`.`	Open water

You may move one cell at a time in the four cardinal directions (up, down, left, right), but not diagonally. You are allowed to revisit cells: any cell other than `#` may be traversed any number of times.

Your route must satisfy all of the following:

- Starting at `A`, the path must end at `B`.
- The path must pass through at least one hydrothermal vent (`^`).
- The path may never pass through rock (`#`).
- The path may never go out of bounds of the grid.

Given a route, can you determine whether it satisfies all the conditions?

Input

The first line of input contains two space-separated integers n and m: the number of rows and columns of the grid, respectively. You are guaranteed that $1 \leq n, m \leq 50$.

The next n lines each contain a string of length m representing one row of the grid. You are guaranteed that the grid contains exactly one `A`, exactly one `B`, and at least one `^`. Each cell is guaranteed to be one of the five types listed above.

The final line contains a string of moves, where each character is one of `U`, `R`, `D`, `L`, meaning up, right, down, and left respectively. Moves are executed in order starting from the `A` on the grid: each character shifts the submarine one cell in the corresponding direction. The string is guaranteed to have length between 1 and 100 inclusive.

Output

Output `YES` if the route satisfies all requirements, and `NO` otherwise.

Sample Input 1

```
5 6
^#A..#
####..
#.....
#.#^^.
.B#..#
RRDDLDDUULLDDLR
```

Sample Output 1

```
YES
```

Sample Input 2

```
3 3
B..
.^.
..A
ULULR
```

Sample Output 2

```
NO
```

Sample Input 3

```
3 4
....
..#.
A^#B
RDRRU
```

Sample Output 3

```
NO
```

Sample Input 4

```
6 3
...
.A.
.#.
.^.
.B.
...
DDD
```

Sample Output 4

```
NO
```

Sample Input 5

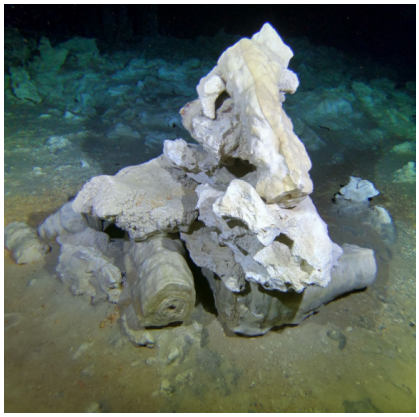
```
3 3
A..
...
^.B
RRDD
```

Sample Output 5

```
NO
```

Problem F
Ariel Collection Management
Time limit: 3 seconds

Ariel Collection Management (ACM) is a firm specializing in maintaining and documenting underwater collections of human-origin objects (HOOs), including gadgets, gizmos, whosits, whatsits, and thingamabobs. ACM has recently been hired to catalog the collection of a high-ranking member of mermaid royalty, who wants to know which HOO she has the most of. You know that the only types of HOOs she has are the five listed above, and that no two types of object have the same count. Can you help ACM figure it out?



Input

The first line of input contains a single integer n ($10 \leq n \leq 2 \cdot 10^5$).

The next n lines of input each contain a single string, either `gadget`, `gizmo`, `whosit`, `whatsit`, or `thingamabob`, representing one of the objects in her collection. You are guaranteed that no two types of object occur the same number of times.

Output

Output two lines.

The first line of output should contain the name of the object type with the highest count (either `gadget`, `gizmo`, `whosit`, `whatsit`, or `thingamabob`).

The second line of output should contain its count, that is, how many times the most common object occurs in the collection.

Sample Input 1	Sample Output 1
10 gadget gizmo whosit whatsit gadget gizmo whosit gadget gizmo gadget	gadget 4

Sample Input 2	Sample Output 2
15 thingamabob whatsit whatsit whosit whosit whosit gadget gadget gadget gadget gizmo gizmo gizmo gizmo gizmo	gizmo 5

Problem G
Ideal Temperature
Time limit: 2 seconds

You're conducting important lab work in your submarine, and you need to make sure that the experiments are happening at the right temperature. Your experimental setup can tolerate small deviations from the ideal temperature, but any deviation that's too large will ruin your data. You've calculated the tolerance precisely, that is, how far from the ideal temperature your setup can withstand. Given the ideal temperature, the tolerance, and the current temperature, can you figure out whether your experiment will be okay?

Input

Input consists of three lines. Each of these lines starts with one of the following strings: `Current`, `Ideal`, or `Tolerance`. After that, an integer x follows $(0 \leq x \leq 100)$. The meaning of x depends on which line it is on:

- On the line starting with Current, this integer x denotes the current temperature c.
- On the line starting with Ideal, it denotes the ideal temperature i.
- On the line starting with Tolerance, it denotes the tolerance t.

These lines may be given in any order.

The initial string and the integer x are space-separated.

Output

Output a single line. If the current temperature c differs from the ideal temperature i by at most the tolerance t, output `YES`. Otherwise, output `NO`.

Sample Input 1	Sample Output 1

`Current 28`
`Ideal 20`
`Tolerance 10`
	`YES`

Sample Input 2	Sample Output 2

`Tolerance 5`
`Ideal 40`
`Current 35`
	`YES`

Sample Input 3	Sample Output 3

`Ideal 20`
`Tolerance 10`
`Current 35`
	`NO`

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Problem H
Treasure Trove
Time limit: 1 second

It's Captain Jack's lucky day! He's been searching all his life for abandoned shipwrecks full of great riches, and his dream has finally come true. In front of him are n treasure chests, each containing varying quantities of gold coins. But having lots of treasure chests of loot to keep track of is too risky: too many opportunities to lose track of all the gold in the stash. So, he's decided that all the gold needs to end up in a single treasure chest. He's not got much time, though, so he needs to consolidate the coins as quickly as possible.

Every second, Captain Jack can move *one* gold coin from any treasure chest to any other treasure chest. Your task is to determine the minimum number of seconds that Captain Jack needs to get all the gold coins into the same treasure chest.

Input

The first line of input consists of a single integer n. You are guaranteed that $1 \leq n \leq 10^5$.

The second line of input consists of n space-separated integers, representing the number of coins in the treasure chests. Each of these integers is guaranteed to be between 0 and 10000, and at least one of them is guaranteed to be nonzero.

Output

Output a single integer, the minimum number of seconds that Captain Jack needs to consolidate all the gold coins into the same treasure chest.

Example

In the first sample test case (shown below), the first treasure chest contains 1 coin, the second chest contains 2 coins, and the third chest contains 7 coins. Captain Jack can consolidate the coins in 3 seconds: he spends 1 second moving the coin in the first treasure chest to the third treasure chest, 1 second moving one of the coins in the second chest to the third, and 1 second moving the other coin in the second chest to the third. It can be shown that 3 seconds is the best he can do in this case.

In the second sample test case (shown below), all the coins are already in the same treasure chest. Therefore it takes him no time to consolidate the coins.

Sample Input 1	Sample Output 1
3 1 2 7	3

Sample Input 2	Sample Output 2
5 0 0 9 0 0	0

Sample Input 3	Sample Output 3
2 3 3	3

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Problem I
Pronounceable Names
Time limit: 3 seconds

You and your commander have discovered a colossal, unexplored swath of the deep sea, and are logging all the new features and landmarks you're coming across. This means you get to name all these features. Unfortunately, your commander is insisting on completely unpronounceable names, such as `XNXAUUO`. No one's going to want to explore a land that they can't even pronounce! As the commander's subordinate, you don't want to change his names too much, but you at least want to make them pronounceable. So, you come up with some rules for what makes a pronounceable string.

A string of Latin letters is *pronounceable* if it never contains three consecutive vowels or three consecutive consonants. The vowels are `A, E, I, O, U`; all other letters are consonants. For simplicity, `Y` will always be considered a consonant.
Given an initial string of uppercase Latin letters, determine the minimum number of single-character replacements needed to make it pronounceable. A single-character replacement changes the letter at one position to a different letter, leaving all other positions unchanged.

Input

Input consists of a single string s of uppercase Latin letters of length n ($1 \leq n \leq 2 \cdot 10^5$).

Output

Output a single integer, the minimum number of characters that need to be changed of s to make it pronounceable.

Sample Input 1	Sample Output 1
`UVA`	`0`

Sample Input 2	Sample Output 2
`CHRISTMAAAAAAS`	`4`

Sample Input 3	Sample Output 3
`XNXAUUO`	`2`

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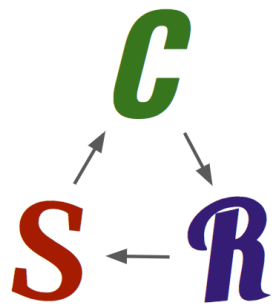
Problem J
Food Not-Chain (Div. 2)
Time limit: 4 seconds

In ecology, interactions between species don't necessarily have to fall into a strict food chain, or even a directed acyclic graph. In fact, we observe a clear cyclical relationship between three particular strains of *Escherichia Coli*: Strain C, a toxin-producing strain; Strain S, a toxin-sensitive strain; and strain R, a toxin-resistant strain.



- Strain C dominates Strain S, because Strain S is vulnerable to the Strain C toxin.
- Strain R dominates Strain C, because it is immune to the toxin.
- Strain S dominates Strain R, because it doesn't have to spend as many resources on toxin resistance.

You wonder to yourself: "Can I find one of these cycles in the wild"? Since there are so many possibilities, you consult your Ecology Specialist to choose just a few interesting species to consider. They decide on a set of n species, labeled 1 through n. They choose these n species very carefully: they ensure that in every pair of different species, one dominates the other. They also ensure that every species among these n dominates at least one other species, guaranteeing that such a 3-species cycle exists.

Your Ecology Specialist has returned with the results of every pair of interactions. Can you find 3 species that form a cycle?

Input

The first line of input contains an integer n ($3 \le n \le 500$).
The next $n(n-1)/2$ lines of input each contain two space-separated integers i, j ($1 \le i, j \le n$, $i \ne j$), indicating that species i dominates species j. You are guaranteed every unordered pair of species $\{i, j\}$ appears exactly once.

Output

Output three space-separated integers a, b, c on a single line, where species a dominates species b, species b dominates species c, and species c dominates species a. It can be shown that a valid triple of species is guaranteed to exist. If multiple valid triples exist, output any one of them.

Sample Input 1	Sample Output 1
4 1 4 3 2 2 4 4 3 1 3 2 1	4 3 2

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Problem K
The Kraken
Time limit: 5 seconds

The Kraken is a legendary, gigantic sea monster of Norwegian folklore, capable of creating whirlpools powerful enough to sink entire vessels. After reading the ancient texts, you've determined that the way to fend off the Kraken is with a zigzag shape, its sharp corners disturbing any semblance of its whirlpools' circular nature. Only a very particular zigzag setup will work, though. Can you create the exact shape you need and save your crew?

Your task is to print a zigzag pattern of width w and height h on a grid of dots (.) and hashes (#). The zigzag pattern you need satisfies the following conditions:

- The top-left cell is a #.
- Each column contains exactly one #.
- A # in the top row has # neighbors diagonally below-left and below-right (provided that these locations stay within grid bounds).
- A # in the bottom row has # neighbors diagonally above-left and above-right (provided that these locations stay within grid bounds).
- All other # cells have exactly one diagonal neighbor above and exactly one diagonal neighbor below.

Input

A single line containing two integers w ($1 \leq w \leq 79$) and h ($2 \leq h \leq 79$).

Output

h lines of exactly w characters each, forming the zigzag pattern.

Sample Input 1	Sample Output 1
12 3	#...#...#... .#.#.#.#.#.# ..#...#...#.

Sample Input 2	Sample Output 2
17 5	#.....#.....# .#.....#.#.....#. ..#...#...#...#.. ...#.#.....#.#... ....#.....#....

Sample Input 3	Sample Output 3
4 7	#... .#.. ..#. ...# ....

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Problem L
The Leviathan
Time limit: 2 seconds

The Leviathan is a mythical dragon-like sea monster symbolizing primordial chaos: impenetrable shields for scales, breath that sets coals ablaze, smoke pouring from its nostrils. Ancient texts describe it as incomprehensibly fearsome, where mere mortals have no hope of even coming close to getting it under control. You and your crew know that you have no chance of slaying the beast, and your only hope is to find a way to avoid it altogether. Luckily, after thousands of years of studying it, you've cracked just a little bit about its movement pattern. Can you use this to determine whether you're safe from its wrath?

You and the Leviathan both occupy cells on an m-by-n grid. Your position is fixed, but the Leviathan may move if it desires. It moves two spaces diagonally: if it starts at cell (r, c), it can move to the cell $(r \pm 2, c \pm 2)$, skipping over the intervening cells, provided that the destination cell is still within bounds of the grid. Your task is to determine whether the Leviathan can make a sequence of zero or more moves in succession to end up at exactly your position.

Input

Input consists of a single line. That line will contain 6 integers: m, n, x_1, y_1, x_2, y_2. The value m denotes the number of rows in the grid, and n denotes the number of columns in the grid. Your position on the grid is given by (x_1, y_1), where rows and columns are 1-indexed; the Leviathan's starting position is given by (x_2, y_2). It is guaranteed that $1 \le x_1, x_2 \le m \le 10^9$, and $1 \le y_1, y_2 \le n \le 10^9$.

Output

Output `YES` if the Leviathan can reach your position in zero or more moves. Otherwise, output `NO`.

Sample Input 1	Sample Output 1
`5 7 1 1 3 7`	`YES`

Sample Input 2	Sample Output 2
`3 3 1 1 1 2`	`NO`

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Problem M
Morse Code
Time limit: 3 seconds

The secret deep sea bunker has intercepted adversary communications! Not only that, they've discovered that they communicate using a series of dots and dashes, a code that they refer to as the "Adversary's Code of Morse" (ACM). They've also cracked the secret encoding, and it's just up to you to put the pieces together. In the ACM, each letter is given a corresponding sequence of dots and dashes, and each secret message consists of just dots, dashes, and Xs. To encode a message, the adversary encodes each letter in order using its corresponding sequence of dots and dashes in the ACM, and then separates consecutive letters' encodings with the character `X`. Using this knowledge, can you decode the secret message?

Input

The first line of input contains an integer n $(1 \leq n \leq 26)$, denoting the number of encoded letters.

The next n lines of input each consist of a single uppercase Latin character, followed by a string of dots (`.`) and dashes (`-`) of length between 1 and 10 inclusive, denoting the encoded version of this character in the ACM. The character and the string will be separated by a single space.

The next line of input contains a string of length at most 1000, containing only dots (`.`), dashes (`-`), and Xs (`X`), denoting the encoded message. It is guaranteed that this string is a valid encoded message under the given encoding.

Output

Output a single string, the decoded message.

Example

In the first sample test case, the encoded version of `S` is `...`, and the encoded version of `O` is `---`. If the original message was `SOS`, it would be encoded as follows: the first `S` becomes `...`, then the `O` becomes `---`, and finally the last `S` becomes `...`. Putting these together, and separating them by `X`s, we get `...X---X...`. Since this matches the provided encoded message, we can deduce that the decoded message is `SOS`.

Sample Input 1	Sample Output 1
2 S ... O --- ...X---X...	SOS

Sample Input 2	Sample Output 2
4 C -.-. H .... P .--. S ... .X...X.--.X-.-.	HSPC

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Problem N
Sport Diving
Time limit: 5 seconds

The Underwater Ventures Association (UVA) has started a new competition: Sport Diving! They've constructed a challenging obstacle course for competitors to race through as fast as possible, to test their aquatic acumen. They've also procured a hefty sum of sponsorship money to distribute to the winners. The only question is: what if there's a tie?

After some intense discussion with UVA, you've decided on the following scheme: UVA wil first decide a payout for each rank (e.g. 1st place = $1,000, 2nd place = $200, 3rd place = $100). Then, if two or more people are tied, they receive the *average* payout for the ranks they're tied for. For example, if two people are tied for 2nd place, they receive the average of the 2nd place and 3rd place payout.

Given the results, can you determine how much money each competitor should win?

Input

The first line of input contains an integer n $(1 \leq n \leq 2 \cdot 10^5)$.

The second line of input contains n nonnegative integers, the payouts for ranks 1 through n in order. All payouts are less than 10^9, and payouts are guaranteed to be non-increasing.

The third line of input contains n positive real numbers, the finishing times of competitors 1 through n. All times have at most 3 decimal places and are less than 10^6.

Output

Output n space-separated real numbers, where the i-th number of output represents the winnings of competitor i. Outputs with an absolute or relative error of at most 10^{-4} will be accepted.

Sample Input 1

```
5
1000 200 100 0 0
5.33 5.54 10.46 5.33 5.54
```

Sample Output 1

```
600.0 50.0 0.0 600.0 50.0
```

Sample Input 2

```
5
1000 200 100 0 0
5.0 4.0 3.0 2.0 1.0
```

Sample Output 2

```
0.0 0.0 100.0 200.0 1000.0
```

Sample Input 3

```
6
999990005 999990004 999990003 999990002 999990001 999990000
999999.999 999999.999 999999.999 999999.999 999999.999 999999.999
```

Sample Output 3

```
999990002.5 999990002.5 999990002.5 999990002.5 999990002.5 999990002.5
```

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